

- In this panel Q&A, we have Dr. Meyer and Dr. Lukaczer with us to address the clinical pearls and pitfalls of stool test interpretation. The first question we're going to ask relates to a situation that we often deal with in our practices where we get a new patient who's coming to us with stool test results from either another practitioner, or now more and more so that they've done themselves through an online ordering platform. So when we get a patient like this, the first question is for what duration of time are those stool tests results valid? Is it a year? Is it a month? Is it somewhere in between? And does it differ for markers of dysbiosis, markers of inflammation, markers of maldigestion and malabsorption? So Nicole, why don't you start off here.

- Thanks, Robert. I think that there's no hard and fast rule here. At least that's my impression of the duration of validity of a stool test. I would probably take it in the context of the patient's concerns. So I would probably consider three main things. I would consider, have the signs and symptoms for the patient's digestive dysfunction changed? Is there anything new or is it the same? Were there any red flags on the original test? Were there markedly elevated levels of say calprotectin? Something like that would give me concern. Was there blood in the stool? And then I would always consider patient preference. Was there a plan associated with this original stool test? Was anything done? Is it affordable for them to do another one? It's a practical answer, but I think an important one when it comes to functional testing because we know it can be expensive.

- Dan, how about you?

- I think those are great thoughts. I would split it up a little bit more, agreeing with everything that Nicole said. And I would say, "Well, what about these digestion and absorption markers? What about these microbiota markers? And what about the inflammation immune dysregulation markers?" So in terms of, you know, how long are they good for? I think Nicole's point is well taken. Do you have the same symptoms? If your symptoms are changed, then maybe that has been part of the change. Have you done anything? Have you taken an antibiotic? Have you taken an anti-protozoal medication because you had blastocystis? Well, if your symptoms change, then, you know, there's a whole nother question. Having said all that, I would say that, you know, in the big picture, I think that a protozoal infection, so talking about a gut microbiota, if there is a positive on that and you haven't done anything, I think that test is good for a while. I think the microbiome is good for, again, in the context of, well, you didn't do much, you just got those tests. It's good for maybe three or six months. Digestion and absorption, I think those are good for three to six months. So in summary, I think the microbiome is, and particularly protozoal infections shorter, and the other ones a little bit longer, but I don't think those other ones are good forever. And again, Nicole's point is well taken. Are you still gonna use them, or are you

just gonna say these are no good for anything? I think you have to take it into context.

- All right, if we have done the stool testing on a patient and they are improving, is a stool test necessary? Are there any clinical scenarios where repeat testing is necessary, either for clinical reasons or for legal reasons? Dan, why don't you start with this one?

- I think it depends, again, but I would say a stool test is necessary in certain situations. I certainly think a repeat stools test is necessary with elevated calprotectin or lactoferrin. I think it's necessary. And I think it's necessary if you find occult blood in the stool, you don't have to do the whole stool test, you can just do occult blood. But those two, I think it's absolutely necessary. And then is it necessary with some of these other things if the patient is improving? I wouldn't say it's absolutely necessary. And in my experience, often if the patient is improving, I don't do it because of the cost. Do I like to do it? Will I do it if I can? I would.

- And Nicole.

- I agree with Dan, especially around the calprotectin, lactoferrin markers. That's not something I tend to leave alone, even if I repeat the test as a standalone. But generally speaking, if some of the findings are more mundane and I see symptom improvement, I don't generally retest with a couple of exceptions. So if there was a bacterial spectrum that was putting the patient at risk for future autoimmune disease, let's say klebsiella pneumoniae, and there was a history of autoimmune disease in the family, I might suggest to a patient that we follow that up in a year or two years. Something like that, because of those pathogenic bacteria. The other situation might be if I'm monitoring a disease state that is heavily dependent on digestion and absorption. So if I'm monitoring a patient potentially with osteoporosis from celiac or something like that, theoretically you could even use it to monitor exposure in a gluten sensitivity with the anti-gliadin antibodies that are present on some of those tests. So there are some situations where I might retest, but not as a hard and fast rule.

- And if there is any guidance you can give around when should you repeat the test, if repeat testing is indicated. Can you give any guidance to the practitioners who are using these tests for the first time?

- I certainly wouldn't retest before six months, Robert, because I think affecting change in the digestive system can sometimes be laborious when there's a multitude of factors at play. If there's a protozoal infection, arguably the retest could be sooner. But when we're trying to shift the microbiome in a positive direction, when we're trying to see an improvement in short-chain fatty acid

production, when we're trying to see inflammatory markers come down if they weren't at critical levels, I would likely retest in about a year. Six months to a year.

– How about you, Dan?

– Similar to what Nicole said. I think three to six months in those particular inflammatory markers, I definitely want to retest those. I think Nicole made a good point about protozoal infections. If you know a patient is getting better, I've done something specific with a protozoal infection, I do kind of want to retest, and that would be three to six months. But I don't always get to do that because they're better. But three to six months in general.

– All right, our last question is really a five-part question. We're gonna ask our subject-matter experts here, our experienced clinicians, what advice they want to give you about pearls and pitfalls in interpreting five different types of biomarkers in the stool test? The maldigestion biomarkers, the inflammation biomarkers, bacterial, parasite, and fungal dysbiosis. So, Dan, why don't you just take this where you will. It's open-ended. When people are new to using this test, what are the pearls and pitfalls for maldigestion markers?

– Well, let me first say in the big picture, all of these things are kind of the pearl I'd like to think is all in context, all putting them together. That doesn't mean that they individually don't have any significance, but I want to see them in context of the other things and in context of that patient. So having said that, hopefully you'll think of that as a pearl. Maldigestion, I would say that I think that I generally do some kind of therapeutic intervention if that pancreatic elastase, or I should say pancreatic elastase specifically is anywhere south of 400 to 350. I think that doesn't necessarily have to be digestive enzymes. It can be chew your food more. It can be use bitters. It can be using something else that increases hydrochloric acid secretion. And by that, you then increase these digestive enzymes. But don't, you know, if it's borderline, I wanna do something. So that's my pitfall, if you will.

– And don't worry, we will be getting to those therapeutic interventions soon in your five R therapeutic intervention training, which is coming up very, very shortly in this course. Nicole, how about you? Anything to add to what Dan has said about maldigestion markers?

– Similar thoughts. I think the message I would give the clinical pearl a little bit here would be think upstream. So a lot of maldigestion starts with patients eating in a setting that's rushed, with not chewing their food, with feeling really stressed when they're eating. And, you know, there's a lot of potential for doing some breathing, for relaxing, anticipating your meal, chewing slowly. And I

have seen that in many cases really alter the digestive tract function, and particularly help people digest their food better. And I love this idea that Dan has of bitters, and even bitter foods, you know, as opposed to herbal bitters. It's just really easy to manipulate the diet in a way that would stimulate better hydrochloric acid production, which in turn, you know, it's this orchestra again, it stimulates pancreatic elastase which stimulates bile, and we go on and on and on. It just goes all the way down through the system if it's working well. So start at the top.

- I always like to say mellow music and mindful mastication for that cephalic phase to help with my maldigestion. Nicole, let's turn to you for the inflammation biomarkers, specifically if you wanna focus on calprotectin. Any pearls and pitfalls for people who are gonna be using this now for the first time in their careers?

- I think it's tempting when you're new to functional medicine and you see an inflammatory marker in any body product, whether that's in serum or in stool, to wanna put out that fire, right? And the bigger question is, what started the fire? So again, what's triggering this inflammation? Is it a protozoal infection? Is it dysbiosis? Is the degree of inflammation relative to what you're seeing on that stool test? Does it really justify, or should you be looking for something else? So I would say, again, think a little bit upstream. Think about what could be driving this inflammation. And if it makes sense on the stool test, if it's well justified, then you can treat accordingly. But trying to put out inflammation by doing anti-inflammatory things in that moment, I would be more focused on looking for a trigger.

- This is the principle we're repeating throughout this course is use some of those therapies as a bridge while you're treating the root causes and then you should be able to wean the patient off those therapies that are really only to be intended as a bridge. Dan, anything you would add to what Nicole said about information?

- I think she made some great points. I would just add that it is, I think, easy enough and often insurance reimbursed to just get a fecal calprotectin. So my pearl, if you will, is just repeat that one test if it's elevated. Repeat it, and obviously after doing some upstream whatever you're finding upstream.

- Excellent point. The calprotectin is widely available with conventional lab companies. Okay, let's combine these last three: dysbiosis, bacterial parasite, fungal biomarkers. What pearls would you provide to a new practitioner new to functional medicine, Dan?

- I think for bacterial and fungal, I would say I would combine them and say that I have over the past 10 years been much more focused on all sorts of different kinds of fibers by crowding out those bacteria as opposed to saying, "I'm going to give you an antimicrobial herb or

an antibiotic." I did a lot more antimicrobial herbs, And there's so many excellent and somewhat specific fibers that you can put into all sorts of smoothies. So I think for bacterial and fungal dysbiosis, I'm not against the idea of killing them, and particularly in fungal, if I think that's a real issue in doing some anti-microbial, anti-fungal. But I really want to crowd them out with specific fibers. For protozoal, I would say the pearl to me actually is that a protozoal infection that I think is causing those symptoms. Or as I said before, or you may or may not have heard this yet, is that, you know, some of these protozoas are benign depending on what species they are.

- Marry the lab result to the patient.

- Yeah.

- And Dan is jumping ahead giving a little spoiler alert for some of the seeding and feeding concepts that you'll be hearing about very shortly in this course. Okay, Nicole, why don't you wrap this Q&A up with your perspective on dysbiosis testing pearls and pitfalls.

- Well, to follow up Dan's point, and one of my big points was don't kill the host. You know, don't kill the host trying to wipe out a bacteria, trying to wipe out protozoa. You know, I think the damage that we can do inadvertently by treating or overtreating some of these situations, even a chronic yeast problem is reasonably significant. And so I love this idea of getting the body on side, getting the immune system on side, crowding out the dysbiotic bacteria by using something like fiber. I think it must our age, Dan, but I'm on the fiber train too. So I really think that that's a gem. You know, there's a lot of terrific fibers out there that can make a massive difference in gut health, and they're underutilized and they've been underutilized. So I would say don't kill the host would be my big takeaway in this section. Try to get the body to work with you so that it will take care of its own problem.

- Sound and sage advice. Thank you, Nicole. Thank you, Dan. We'll continue now with the course.